

Student Guide — primary / module-07-eleven-multiplication

IKS Curriculum

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Student Workbook

Student Workbook — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Ekādhikena Pūrveṇa* — _____
- *sūtra* — _____
- *Anurūpyena* — _____

Day-by-day notes

Day 1 — The Hook — What Is the Trick?

What I learned:

One question I have:

Day 2 — The Carry Case

What I learned:

One question I have:

Day 3 — Why It Works — The Algebra

What I learned:

One question I have:

Day 4 — 3-Digit $\times$ 11

What I learned:

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Day 5 — Speed Day

What I learned:

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Day 6 — Anurūpyena — $\times 12$ to $\times 19$

What I learned:

One question I have:

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What I learned:

One question I have:

Day 8 — Tirthaji and Historical Context

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Hook — What Is the Trick?

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “What’s one thing you remember from yesterday?”

Today’s idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthajī”)
- Watch this trick! $23 \times 11 = ?$

Show on the board: “2 _ 3” with a gap. Whisper: “What’s 2 + 3?” (Wait for “5”.) Fill in: “2 5 3 → 253!”

Each child practices: $12 \times 11 = 132$. $34 \times 11 = 374$. $25 \times 11 = 275$. Try 5 more.

Magic chant: “*Split, add, put in middle!*” (Children chant 3 times.)

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Carry Case

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 87×11 , 65×11 correctly using the carry rule.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)
- What if the middle sum is too big? Like 87×11 .

Show: “8 _ 7” with gap. $8 + 7 = 15$. Whisper: “15 is two digits! Keep the 5, carry the 1.”

Result: $8 + 1 = 9$, then 5, then 7 → **957**.

Practice: $67 \times 11 = 7\ 3\ 7$. $95 \times 11 = 1\ 0\ 4\ 5$. “*Carry the tens, keep the ones!*” (Chant.)

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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Day 3 — Why It Works — The Algebra

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Derive $(10a+b)\times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)
- Why does the magic work?

Show 23×11 with paper strips: - One strip of 23 + ten strips of 23 = 11 strips. - Stack them: $230 + 23 = 253$. - The “230” gives the 2 and the 3 (separated). The “23” gives the +2 in the tens and +3 in the ones. - $2 + 3 = 5$ in the tens place. The 2 and 3 stay on the ends!

Magic? No — just adding! Children try with strips of 14: $14 \times 11 = 140 + 14 = 154$.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

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Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)
- For 3-digit numbers: split with TWO gaps!

$234 \times 11 = ?$ - Write “2 _ _ 4” with two gaps. - Fill: $2+3 = 5$ (first gap), $3+4 = 7$ (second gap). - Answer: **2574**.

Practice: $123 \times 11 = 1 (1+2) (2+3) 3 = 1353$. $456 \times 11 = 5016$ (with carries — hard).

Most children stick to no-carry cases this day.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 5 — Speed Day

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Complete a 50-problem mental-math sprint in 5 minutes.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)
- Speed time! Today we play a quick ×11 game.

In pairs: one child says a 2-digit number, the other shouts the answer × 11. Switch every 30 seconds. 5 rounds.

End with class cheer: “*Split! Add! Middle! Done!*”

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)
- We can do $\times 12$ too! Just split into $\times 10 + \times 2$.
- $15 \times 12 = (15 \times 10) + (15 \times 2) = 150 + 30 = \mathbf{180}$.

Practice in pairs: one says a number, the other doubles it for $\times 2$, then adds a zero for $\times 10$, then adds the two together for $\times 12$.

Try $\times 13$, $\times 14$ only if time. Most children stick to $\times 12$ today.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

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Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Recognise when traditional long multiplication is faster than the sūtra method.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tīrthajī”)

- The trick is special! It works best for $\times 11$ (and $\times 12$).
- For things like $\times 23$ or $\times 47$, the trick is hard. Then we use the regular way.

Show: 23×47 . “*Would the trick work?*” (No — 47 is too far from 10.) “*So we do it the long way.*”

Children practice both methods on 24×11 vs 24×23 to feel the difference.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- If energy is low, swap the activity for free drawing.

Day 8 — Tirthaji and Historical Context

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Explain who Tirthaji was and why scholars debate the “Vedic” attribution.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)
- The trick comes from a book by Bharati Krishna Tirthaji, who lived about 100 years ago.
- He named it after old Sanskrit ideas. Today, scholars are not sure if the methods are really thousands of years old. But the *trick still works* — and that's what matters for us!

Show a photo (or drawing) of Tirthaji. Children put a thumbs-up if they like the trick.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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Day 9 — Mixed Practice

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Solve a mixed problem set using best-fit method per problem.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions)
- Today we play a sorting game!

Show 10 problems on cards. For each, children hold up a sign: - “Trick!” (for $\times 11$ problems) - “Long!” (for harder problems)

Then solve 5 $\times 11$ problems for practice. End with applause.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Take the summative quiz and reflect on which method works best.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “What's one thing you remember from yesterday?”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthajī”)
- Final day! We celebrate the $\times 11$ trick.

Each child shows their best $\times 11$ problem to a partner. Cheer for each one.

Take the small quiz with stickers (every child gets a “Number Magician” sticker).

End: class chant — “Split! Add! Middle! Magic!”

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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Homework

Homework — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Draw about today's concept. (~5 min)
2. **Day 2 — The Carry Case** · Draw about today's concept. (~5 min)
3. **Day 3 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
4. **Day 4 — 3-Digit $\times 11$** · Draw about today's concept. (~5 min)
5. **Day 5 — Speed Day** · Draw about today's concept. (~5 min)
6. **Day 6 — Anurūpyena — $\times 12$ to $\times 19$** · Draw about today's concept. (~5 min)

7. **Day 7 — Limits of the Method** · Draw about today's concept. (~5 min)
8. **Day 8 — Tirthaji and Historical Context** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.